Smoke test for `math.sty`

This file exercises all commands in `math.sty` to check that the style works out of the box.

Basic brackets

$$\{x^2+y\}, \quad (x^2+y), \quad f(x^2+y), \quad [x^2+y], \quad \langle x^2+y\rangle$$

Other brackets

$$|x_t+y^2|, \|x_t+y^2\|, \lfloor x_t+y^2\rfloor, \lceil x_t+y^2\rceil \ldots$$

Functions and operators

$$\ln x, \ln(x), \exp x, \exp(x), \mathbb{1}, \mathbb{1}_A(x), \mathbb{1}(x), \operatorname{tr}, \operatorname{tr}(M)$$

$$\operatorname{var},\ \operatorname{var}(X),\ \operatorname{var}_t(X), \quad \operatorname{cov},\ \operatorname{cov}(X,Y),\ \operatorname{cov}_t(X,Y)$$

$$\operatorname{corr},\ \operatorname{corr}(X,Y),\ \operatorname{corr}_t(X,Y), \quad \operatorname{sd},\ \operatorname{sd}(X),\ \operatorname{sd}_t(X), \quad \operatorname{se},\ \operatorname{se}(X),\ \operatorname{se}_t(X)$$

$$\max,\ \max\{x,y^2\},\ \max(x,y^2),\ \max_n(y_n^2), \quad \min,\ \min(x,y^2),\ \min_n(y_n^2)$$

$$\sup,\ \sup(S),\ \sup_n(y_n), \quad \inf,\ \inf(S),\ \inf_n(y_n)$$

$$\operatorname*{argmax}_{x\in\mathcal{X}} f(x), \quad \operatorname*{argmin}_{x\in\mathcal{X}} f(x)$$

$$\mathbb{E},\ \mathbb{E}_t,\ \mathbb{E}(X^2),\ \mathbb{E}_t(X_t \mid B), \quad \mathbb{P},\ \mathbb{P}_t,\ \mathbb{P}(A \mid B),\ \mathbb{P}_t(A_t \mid B)$$

Derivatives

$$\frac{dy}{dx},\ \frac{d^n y}{dx^n},\ dy/dx,\ d^n y/dx^n$$

$$\frac{\partial y}{\partial x},\ \frac{\partial^n y}{\partial x^n},\ \left.\frac{\partial y}{\partial x}\right|_z, \quad \partial y/\partial x,\ \partial^n y/\partial x^n,\ \partial y/\partial x|_z$$

$$\frac{d\ln(y)}{d\ln(x)},\ d\ln(y)/d\ln(x),\ \frac{\partial\ln(y)}{\partial\ln(x)},\ \left.\frac{\partial\ln(y)}{\partial\ln(x)}\right|_z,\ \partial\ln(y)/\partial\ln(x),\ \partial\ln(y)/\partial\ln(x)|_z$$

Logical and statistical relations

$$A \Leftrightarrow B, \qquad A \Rightarrow B$$

$$X_n \overset{iid}{\sim} \mathcal{N}(0,1), \quad X_n \xrightarrow{as} X, \quad X_n \xrightarrow{p} X, \quad X_n \xrightarrow{d} X$$

$$\operatorname*{ess\,sup}_{x\in\mathcal{S}} f(x), \qquad \operatorname*{ess\,inf}_{x\in\mathcal{S}} f(x)$$

Accents

$$\underline{x(t)},\ \overrightarrow{x(t)},\ \overline{x(t)},\ \widehat{x(t)},\ \widetilde{x(t)}$$

Complex parts

$$\mathrm{Re}(z),\ \mathrm{Im}(z)$$

Greek letters

$$\alpha, \beta, \chi, \delta, \epsilon, \varepsilon, \phi, \varphi, \gamma, \eta, \iota, \kappa, \lambda, \mu, \nu, \omega, \varpi, \psi, \rho, \sigma, \theta, \vartheta, \xi, \zeta$$
$$\Delta, \Phi, \Gamma, \Lambda, \Omega, \Sigma, \Theta, \Upsilon, \Xi$$

Blackboard-bold letters

$$\mathbb{R},\ \mathbb{N},\ \mathbb{Z},\ \mathbb{Q},\ \mathbb{C}$$

Bold letters

$$\mathbf{q},\ \mathbf{u},\ \mathbf{v},\ \mathbf{w},\ \mathbf{y}$$
$$\mathbf{A},\ \mathbf{B},\ \mathbf{H},\ \mathbf{J},\ \mathbf{K},\ \mathbf{M},\ \mathbf{V},\ \mathbf{W},\ \mathbf{Y}$$

Calligraphic letters

$$\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{F}, \mathcal{G}, \mathcal{H}, \mathcal{I}, \mathcal{J}, \mathcal{K}, \mathcal{L}, \mathcal{M}, \mathcal{N}, \mathcal{O}, \mathcal{P}, \mathcal{Q}, \mathcal{R}, \mathcal{S}, \mathcal{T}, \mathcal{U}, \mathcal{V}, \mathcal{W}, \mathcal{X}, \mathcal{Y}, \mathcal{Z}$$